

SOFTWARE ENGINEERING LAB

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #05 | Campus & Academic Operations

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1. Problem Context & Overview

University departments need an automated elective course selection mechanism where students allocate bidding credits across preferences and the engine solves allocation constraints while preventing timetable collisions.

Target Stakeholders / Actors: Student, Academic Registrar, Academic Timetable System.

2. Requirements Table

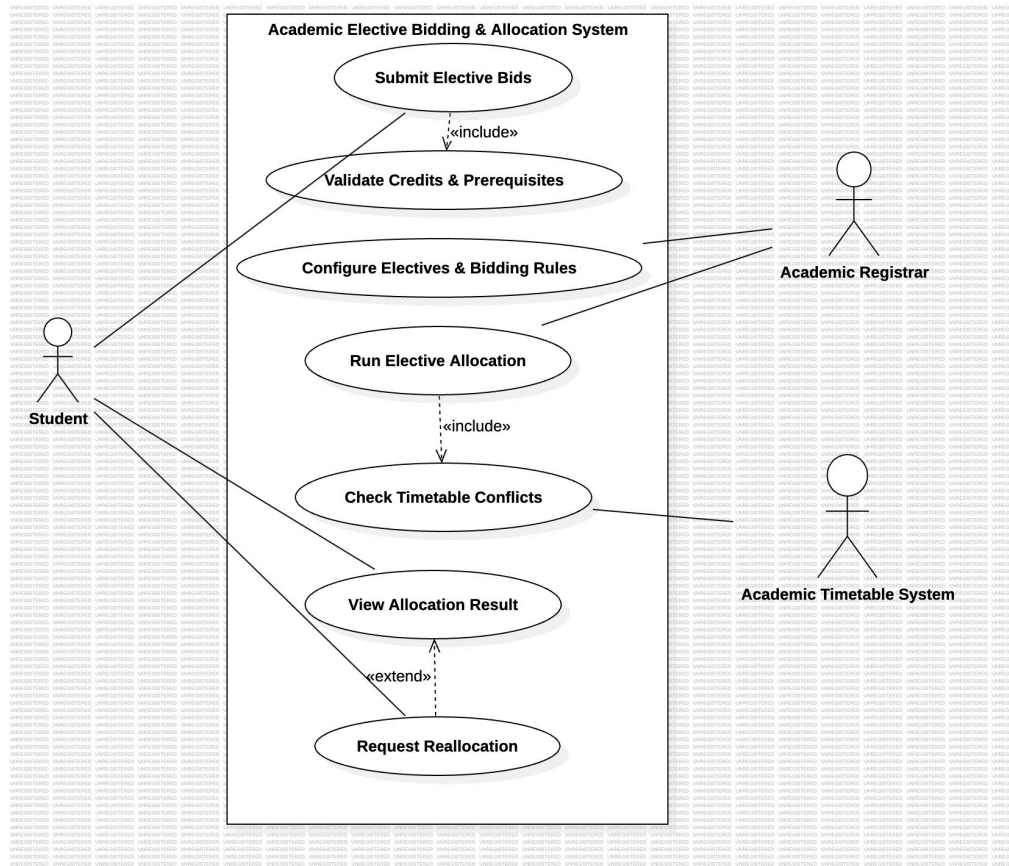
2.1 Functional Requirements (FR-001 to FR-005) and Non-Functional Requirements (NFR-001, NFR-002)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions before a bid is accepted.	High	Pass: Credits deduct accurately and prerequisites are validated. Fail: Total allocated credits exceed 100 or an unmet prerequisite is bypassed.	Ensures fair credit-based prioritisation and prevents students from bidding for electives they are not academically eligible to take.
FR-002	Functional	The system shall allow the Academic Registrar to configure the list of offered electives, seat capacity per elective, and the bidding window (open/close dates).	High	Pass: Configured electives, capacities, and bidding dates are saved and enforced. Fail: Bids are accepted outside the configured window or exceed seat capacity limits.	Gives the Registrar administrative control over the electives on offer and the rules governing the bidding cycle.
FR-003	Functional	The system shall run an elective allocation algorithm that assigns electives to students based on bid credit ranking and available seat capacity once the bidding window closes.	High	Pass: Every student is allocated the highest-ranked elective their credits and seat availability allow. Fail: A student with a higher valid bid is not prioritised over one with a lower bid for the same seat.	Automates a fair and consistent allocation process that would otherwise be error-prone if done manually.
FR-004	Functional	The system shall check each candidate allocation against the Academic Timetable System to detect timetable clashes before finalising the result.	High	Pass: No student is allocated two electives with overlapping time slots. Fail: A finalised allocation contains a timetable collision.	Prevents students from being allocated electives that they cannot physically attend due to schedule conflicts.
FR-005	Functional	The system shall allow students to view their final allocation result and submit a	Medium	Pass: Students can view results and, where applicable, submit a reallocation request	Provides transparency to students and a corrective path for

ID	Type	Description	Priority	Acceptance Criteria	Rationale
		reallocation request if their allocated elective is unsatisfactory or a conflict is flagged.		within the allowed window. Fail: Results are inaccessible or a reallocation request cannot be raised.	unresolved conflicts or unsatisfactory allocations.
NFR-001	Performance & Security	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the allocation engine scales to the full student population without delaying result publication.
NFR-002	Usability & Reliability	The system shall remain available with at least 99.5% uptime during the active bidding window and shall render the allocation result page within 3 seconds for up to 2,000 concurrent students.	Medium	Pass: Uptime and page-load logs during the bidding window meet the stated thresholds. Fail: Downtime exceeds the allowed threshold or result pages take longer than 3 seconds to load.	The bidding window is time-bound, so the system must stay reliably accessible and responsive to avoid students losing the opportunity to bid or view results.

3. UML Use-Case Diagram

The diagram below models the Academic Elective Bidding & Allocation System, showing all actors (Student, Academic Registrar, Academic Timetable System), the primary use cases, one «include» relationship (Run Elective Allocation includes Check Timetable Conflicts; Submit Elective Bids includes Validate Credits & Prerequisites), and one «extend» relationship (Request Reallocation extends View Allocation Result).



4. Use-Case Flow Specification

Core Use Case: Run Elective Allocation

Use Case Name	Run Elective Allocation
Actors	Academic Registrar (primary), Academic Timetable System (supporting), Student (beneficiary)
Preconditions	1. The bidding window has closed and no further elective bids can be submitted. 2. All submitted bids have already passed the Validate Credits & Prerequisites check. 3. Electives, seat capacities, and bidding rules have been configured by the Academic Registrar.
Postconditions	1. Every eligible student has a finalised elective allocation (or is flagged for reallocation). 2. No finalised allocation contains a timetable collision. 3. Allocation results are persisted and made available for students to view.

Main Success Scenario

1. The Academic Registrar triggers the elective allocation process after the bidding window closes.
2. The system retrieves all validated student bids along with the configured elective list and seat capacities.
3. The system includes the Check Timetable Conflicts use case to verify that each candidate elective slot does not clash with the student's existing timetable.
4. The system ranks students for each elective based on the number of bidding credits committed, highest first.
5. The system allocates electives to students in ranked order until seat capacity for each elective is exhausted.
6. The system stores the finalised allocation results against each student record.
7. The system notifies students that their allocation results are available and enables the View Allocation Result use case.

Alternate Flow: Timetable Conflict Detected

At step 3, if the Check Timetable Conflicts use case detects that a candidate elective slot overlaps with a course already locked in the student's timetable:

- 3a.1 The system rejects that specific elective for the student and does not deduct the associated bidding credits.
- 3a.2 The system automatically considers the student's next-ranked elective preference in their remaining bid pool.
- 3a.3 The use case resumes at main step 4 with the updated bid pool.
- 3a.4 If no conflict-free elective remains, the student's record is flagged as "Unallocated" and made eligible for the Request Reallocation extension.